Séminaire Lotharingien de Combinatoire **XX** (2027)
Article #YY, 14 pp.

Proceedings of the 39th Conference on Formal Power Series and Algebraic Combinatorics (Galway)

The $(s-1)$-adic valuation of Hikita's $\star$-product: a block law, cumulant leads, and a box symmetry

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Abstract. Hikita introduced a two-parameter deformation $\star$ of the product of symmetric functions. Writing $e_{\lambda_1} \star \cdots \star e_{\lambda_\ell} = \sum_\mu c_{\lambda\mu}(s,t)\, e_\mu$, we show that the $(s-1)$-adic valuation of $c_{\lambda\mu}$ is $\ell(\lambda) - \kappa(\lambda,\mu)$, where κ is the maximal number of blocks into which λ and μ split compatibly. Leading coefficients factor over block decompositions into connected pieces. For a full merge the connected piece is a connected-graph cumulant with edge weights $t^{\lambda_i\lambda_j} - 1$. It interpolates between partition-lattice Möbius values at $t = 0$ and weighted Cayley counts at $t = 1$, and a gauge-invariant ratio separates it from Dołęga's column cumulants for $t \neq 0$. The coefficients are invariant, up to a power of s, under complementation in a box. All second-order leading coefficients are given in closed form, via a residue formula for Green polynomials at two-part classes.

Keywords: Hikita star product, Hall–Littlewood functions, elementary symmetric functions, cumulants, valuations

1 Introduction

Hikita [9] defined a commutative, associative product $\star$ on symmetric functions over $\mathbb{Q}(q,t)$ by transporting the ordinary product through the level-one polynomial representation of the affine Hecke algebra; we always say *Hikita's* $\star$. We write $s := q^{-1}$. At $s = 1$, $\star$ is the ordinary product, and at $s = 0$ (after rescaling) it degenerates to a Hall–Littlewood product (Remark 2.3). Di Francesco and Kedem [4, §8.3] observed that the associated t-deformed graded characters lose Schur positivity at generic t, and positivity of $\star$-structure constants fails in every natural basis we tried. *Positivity is lost, but structure is not.* This abstract shows that the structure survives in the $(s-1)$-adic filtration.

We expand the structure constants of $e^\star_\lambda := e_{\lambda_1} \star \cdots \star e_{\lambda_\ell} = \sum_\mu c_{\lambda\mu}(s,t)\, e_\mu$ around $s = 1$. We start from Hikita's formula for $e_k \star (\cdot)$ as a sum over k-subsets (Proposition 2.1). In it, s enters only through an Euler operator, so the $(s-1)^p$ Taylor piece of $e_k \star (\cdot)$ is a differential operator of order p, which can glue at most p existing blocks of parts to a new part. This bounds the valuation below; exactness is settled at $t = 0$, where $e^\star_\lambda$ is a modified Hall–Littlewood function and nothing cancels. Main results:

*[**TODO:** Robin: authorship, presenter and affiliation are your decision (asks sent 2026-10-05, no reply yet). The mathematics in this draft was produced by an AI agent; see the AI declaration.]

- **Block valuation law** (Theorem 3.4): $v_{s-1}(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda,\mu)$ for all $\mu \trianglerighteq \lambda$, $\mu \neq \lambda$, where κ is the maximal number of compatible blocks. The first-order term is a symmetric biderivation with an explicit matrix (Theorem 3.2 and Proposition 3.3).
- **Merge weights and full-merge leads** (Theorems 3.8 and 4.1): the leading coefficient for $\mu = (n)$ is $\frac{1-t^n}{\prod_i(1-t^{\lambda_i})} \sum_{H \text{ connected}} \prod_{ij\in H}(t^{\lambda_i\lambda_j} - 1)$. The graph identity is due to Dołęga [5]; the identification with $\star$ is ours.
- **Block multiplicativity** (Theorems 4.4 and 4.5): every lead is a sum over block decompositions of products of connected leads (exp of connected).
- **Box Complement** (Theorem 5.1): $c_{N^\ell-\lambda,\,N^\ell-\mu} = s^{N\binom{\ell}{2}-(\ell-1)|\lambda|}\, c_{\lambda\mu}$, and a Column Lemma.
- **Second order** (Theorem 6.6 and Corollary 6.7): every valuation-2 lead is in closed form, via a constant-term formula and two-part Green polynomials [7, 2, 13].

At valuation 1 the leads are governed by pairing with p_n, evaluated by a principal specialisation, i.e. one t-string $1, t, \dots, t^{k-1}$ (Lemma 3.9); at valuation 2 by pairing with $p_x p_y$, where the poles of a constant-term integral fall into two interacting t-strings (Theorem 6.3). The full-merge lead is a cumulant: up to a prefactor it is the connected part of the character $e_k \mapsto t^{\binom{k}{2}}$. This places $\star$ next to the Macdonald cumulants of Dołęga [5] and Haglund–Tewari [8]. All three share the graph half (Theorem 4.1 and the remark after it), but a gauge-invariant ratio separates $\star$ from Dołęga's $\oplus$-cumulants away from $t = 0$.

Proofs are sketched here; full proofs will appear in a longer version.

2 Background: Hikita's product and Hall–Littlewood notation

Notation. $\Lambda = \mathbb{Q}(s,t)[e_1, e_2, \dots]$; $I = (e_1, e_2, \dots)$ is the augmentation ideal, so $I^b = \operatorname{span}\{e_\mu : \ell(\mu) \geq b\}$. For G homogeneous of degree n, $\operatorname{lin}_e G := [e_n]G$. $[m]_q = (1-q^m)/(1-q)$ and $[m] = [m]_t$, $\varphi_m = \prod_{i=1}^m (1-t^i)$, $n(\lambda) = \sum_i (i-1)\lambda_i$, $\trianglerighteq$ is dominance. $P_\lambda(x;t)$, $Q_\lambda = b_\lambda P_\lambda$, Q'_λ are the Hall–Littlewood functions of [13, Ch. III], with the Hall pairing $\langle\cdot,\cdot\rangle$ and the HL pairing $\langle p_\lambda, p_\mu\rangle_t = \delta_{\lambda\mu} z_\lambda \prod_i (1-t^{\lambda_i})^{-1}$.

Hikita's product. $F \star G := \mathfrak{q}(\mathfrak{q}^{-1}F \cdot \mathfrak{q}^{-1}G)$, where $\mathfrak{q}$ is the level-one polynomial representation map $F(Y) \mapsto F(Y) \bullet 1$ of the affine Hecke algebra of GL_m. By [9, Lemma 3.1, Cor. 3.9], $\mathfrak{q}$ is a $\mathbb{Q}(q,t)$-linear isomorphism, compatible with $m \to m-1$, so $\star$ is well defined and restricts to Λ [9, Def. 3.4, Lemma 3.2, Cor. 3.10]. All structure constants below lie in $\mathbb{Q}[s,t]$ (Theorem 2.2); specialisations in s are specialisations of these. We use $s = q^{-1}$; then Hikita's Pieri rule [9, Thm. 3.12] reads $e_1 \star e_r = (1-s)[r+1]e_{r+1} + s\, e_1 e_r$.

Proposition 2.1 (Subset formula). *For $F \in \Lambda$ and every N, in $\Lambda_N = \mathbb{Q}(s,t)[x_1,\dots,x_N]^{S_N}$*

$$e_k \star F = E_k F := \sum_{|A|=k} c_A\, X_A\, F(X_{A^c}, sX_A), \qquad c_A = \prod_{i\in A,\, j\notin A} \frac{x_i - t x_j}{x_i - x_j}, \quad X_A = \prod_{i\in A} x_i.$$

Proof idea. By [9, Def. 3.4, Lemma 3.3], $e_k \star F = t^{-\binom{k}{2}} e_k(Y) \bullet F$, where $Y_1,\dots,Y_N$ are the Cherednik operators. Order each monomial of $e_k(Y)$ by increasing index, and put $\pi F = x_1 F(x_2,\dots,x_N, s x_1)$. A Coxeter length count on minimal coset representatives of $S_k \times S_{N-k}$ gives $e_k(Y)F = t^{\binom{k}{2}} \sum_{|D|=k} T_{w^D}\, \pi^k F$. For G symmetric in the first k and the last $N-k$ variables, the chain factorisation of this coset sum and the Poincaré identity $\sum_{w\in S_k} w\big(\prod_{i<j} \frac{x_i - t x_j}{x_i - x_j}\big) = [k]_t!$ give $\sum_D T_{w^D} G = \sum_{|A|=k} c_A\, G^{(A)}$, where $G^{(A)}$ places the head variables on A. □

Write $e^\star_\lambda := E_{\lambda_1} \cdots E_{\lambda_\ell}(1) = \sum_\mu c_{\lambda\mu}(s,t)\, e_\mu$. The variable s enters only through $F(X_{A^c}, sX_A) = s^{\Delta_A} F$, $\Delta_A = \sum_{i\in A} x_i \partial_i$; Taylor expansion at $s = 1$ gives

$$E_k = \sum_{p\ge 0} (s-1)^p E_k^{(p)}, \qquad E_k^{(p)} = \sum_{|A|=k} c_A X_A \binom{\Delta_A}{p}, \tag{2.1}$$

$E_k^{(p)}$ is a differential operator of order $\le p$, with $E_k^{(0)} = e_k \cdot$ and $E_k^{(p)}(1) = 0$ for $p \ge 1$.

Theorem 2.2 (Dominance support; the $s = 0$ valuation). *$e^\star_\lambda = s^{n(\lambda)} e_\lambda + \sum_{\mu \rhd \lambda} c_{\lambda\mu} e_\mu$ with $c_{\lambda\mu} \in \mathbb{Q}[s,t]$ and $c_{\lambda\mu}|_{s=1} = 0$. Moreover $c_{\lambda\mu} \neq 0$ for every $\mu \trianglerighteq \lambda$, and the s-adic valuation of $c_{\lambda\mu}$ is exactly $n(\mu)$. At $t = 0$ the s-leading coefficients $[s^{n(\mu)}]c_{\lambda\mu}$ are all 1: in the basis $b_\mu = s^{n(\mu)} e_\mu$, $e^\star_\lambda$ at $s = t = 0$ is the zeta function of dominance order.*

Proof idea. A termwise degree count on the subset formula: for symmetric P let $\delta_j(P)$ be the maximal total degree in j variables; E_k raises δ_j by at most $\min(k,j)$, and $\delta_j(P) \le \sum_i \min(\rho_i, j)$ for all j characterises $\operatorname{span}\{e_\nu : \nu \trianglerighteq \rho\}$ [13, I (1.11), (6.6)]. □

Remark 2.3 (The $s \to 0$ edge; no novelty claim for the matrix). Since $s = q^{-1}$, the edge $s \to 0$ is Hikita's $q \to \infty$, where the unrescaled product degenerates [9, Lemma 6.3]. In the basis b_μ, $e_k \star (\cdot)$ preserves $\bigoplus_\mu \mathbb{Q}[s,t]\, b_\mu$, and modulo s it is conjugate, via $b_\mu \mapsto t^{-n(\mu')} P_{\mu'}(x; t^{-1})$, to multiplication by $t^{-\binom{k}{2}} e_k$. Consequently the leading coefficients $d_{\lambda\mu}(t) := [s^{n(\mu)}]\, c_{\lambda\mu}$ of Theorem 2.2 *are the known $e \to$HL transition matrix* of [12, §3.2]: writing $e_\lambda = \sum_\kappa M(e,P)_{\lambda\kappa} P_\kappa(x;t)$,

$$d_{\lambda\mu}(t) = t^{\,n(\mu') - n(\lambda')}\, M(e,P)_{\lambda\mu'}(t^{-1}).$$

Their positivity and $d_{\lambda\mu}(1) = \#\{\text{0-1 matrices with row sums } \lambda, \text{ column sums } \mu'\}$ are therefore classical [12, §3.2]; what is new is only the identification of this matrix with the s-leading coefficients of $\star$.

3 The block valuation law at $s = 1$

Definition 3.1. For $\lambda, \mu \vdash n$, $\kappa(\lambda, \mu)$ is the largest r such that $\lambda = \lambda^1 \sqcup \cdots \sqcup \lambda^r$ and $\mu = \mu^1 \sqcup \cdots \sqcup \mu^r$ as multisets of parts, with every block nonempty and $\mu^i \trianglerighteq \lambda^i$ (so $|\mu^i| = |\lambda^i|$); $\kappa = -\infty$ if $\mu \not\trianglerighteq \lambda$. μ is a *coarsening* of λ if $\kappa(\lambda, \mu) = \ell(\mu)$. v denotes the $(s-1)$-adic valuation.

Theorem 3.2 (First order is a biderivation)**.** *Let $D_k := E_k^{(1)}$ and $B(f, g) := \partial_s(f \star g)|_{s=1}$. Then D_k is a derivation of Λ_N, and*

$$B(f, g) = \sum_{k,l} M_{kl} \frac{\partial f}{\partial e_k} \frac{\partial g}{\partial e_l}, \qquad M_{kl} := D_k(e_l) = B(e_k, e_l) = M_{lk}.$$

So B is a symmetric biderivation, $B(f, g) = L(fg) - fL(g) - gL(f)$ for $L = \frac{1}{2} \sum M_{kl} \partial^2 / \partial e_k \partial e_l$.

Proof idea. $\binom{\Delta_A}{1} = \Delta_A$ is a derivation and left multiplication by $c_A X_A$ preserves this. Write $f = \sum_\rho a_\rho(s) e_\rho^\star$, differentiate $f \star g = \sum_\rho a_\rho(s) E_\rho g$ at $s = 1$, and use $D_k(1) = 0$. Symmetry of M is commutativity of $\star$ [9]. □

Proposition 3.3 (The matrix M)**.** *For $k \geq r \geq 0$, with $L(a, b) := \dfrac{(1 - t^{a+b})(t^{ab} - 1)}{(1 - t^a)(1 - t^b)}$,*

$$M_{kr} = B(e_k, e_r) = r\, e_k e_r + \sum_{j=1}^{r} L(k - r + j,\, j)\, e_{k+j} e_{r-j}.$$

Consequently $[(s-1)^1] e_\lambda^\star = \sum_{i<j} M_{\lambda_i \lambda_j} e_{\lambda \setminus \{\lambda_i, \lambda_j\}}$.

Proof idea. Differentiate Corollary 5.5 (whose proof does not use Proposition 3.3) at $s = 1$. Only $y = r$ survives at $s = 1$ ($\pi_a(b)|_{s=1} = \delta_{b0}$); its derivative is $r\, e_k e_r$. For $y = r - j$, $j \geq 1$, put $a = k - r + j$. Then $\partial_s \pi_a(j)|_{s=1} = [u^j] \sum_{m<a} \frac{ut^m}{1+ut^m} = (-1)^{j-1} [a]_{tj}$, so $\partial_s C_{a,j}|_{s=1} = -\frac{[a+j]}{[a]} [a]_{tj} = L(a, j)$. The rest follows from Theorem 3.2, since $D_k(1) = 0$. □

Theorem 3.4 (Block valuation law)**.** *For all $\lambda, \mu \vdash n$ and all t, $v(c_{\lambda\mu}) \geq \ell(\lambda) - \kappa(\lambda, \mu)$. For $\mu \trianglerighteq \lambda$, $\mu \neq \lambda$, equality holds over $\mathbb{Q}(t)$ and at $t = 0$:*

$$v(c_{\lambda\mu}) = \ell(\lambda) - \kappa(\lambda, \mu).$$

At $t = 0$ the leading coefficient is $(-1)^{\ell - \kappa} N(\lambda, \mu)$ where $N(\lambda, \mu) \geq 1$ counts $(k_{ij})_{i<j}$ in $\mathbb{Z}_{\geq 0}$, exactly $\ell - \kappa$ nonzero, with $\lambda + \sum k_{ij}(\epsilon_i - \epsilon_j)$ a rearrangement of μ (ϵ_i the unit vectors).

Proof idea. Lower bound (subset formula only): by polarization, an order-p operator glues at most p existing "blocks" to the new part, so the $(s-1)^m$ coefficient is a sum of products over $\geq \ell - m$ blocks, each of type λ_C by the degree count of Theorem 2.2.

Equality: at $t = 0$, $\omega e^\star_\lambda = \widetilde{H}_\lambda(x;s) = s^{n(\lambda)} Q'_\lambda(x;s^{-1})$, and Macdonald's raising operators $Q'_\lambda(x;u) = \prod_{i<j} \frac{1-R_{ij}}{1-uR_{ij}} h_\lambda$ [13, III (2.15′) with §5 Ex. 7(a)] give R^k_{ij} the coefficient $(u-1)u^{k-1} = -(s-1)s^{-k}$ at $u = s^{-1}$. So every transfer edge costs exactly one $(s-1)$ with sign -1, so nothing cancels; the minimum edge count is $\ell - \kappa$ (components $\le \kappa$; a path tree on each indecomposable block). Specialise $t = 0 \to$ generic. □

Remark 3.5 (Input at $t = 0$). The $t = 0$ edge $e^\star_\mu|_{t=0} = s^{n(\mu)} P_{\mu'}(x; 1/s, 0) = \omega \widetilde{H}_\mu(x;s)$ is **not ours**: it is assembled from [3] eq. (5.15) (with $M_{k,1} = E_k|_{t=0}$), Cor. 5.8 in its level-one form (5.25), Cor. 5.18 (eq. (5.27)) (numbering of arXiv:1505.01657v2), and [13, VI (5.1), (4.14)(iv)]. So Theorem 3.4 does not depend on Cherednik's nonsymmetric theory.

Remark 3.6 (Novelty). We have not found the block law stated anywhere. Its $t = 0$ case is classical in substance: the raising-operator formula follows from [2, eq. (11)] and Jacobi–Trudi, after which the valuation count above is an exercise. New are the lower bound for all t (subset formula alone), the exactness over $\mathbb{Q}(t)$, and the merge weights below.

Theorem 3.7 (Coarsenings: merge-history expansion)**.** *Let $\mu \neq \lambda$ be a coarsening of λ and $m = \ell(\lambda) - \ell(\mu)$. Then $v(c_{\lambda\mu}) = m$, and* $\mathrm{Lead}_{\lambda\mu}(t) := [(s-1)^m] c_{\lambda\mu}$ *equals*

$$\mathrm{Lead}_{\lambda\mu}(t) = \sum_{\text{tight histories } H \text{ ending at } \mu} \ \prod_{\text{merges in } H} W_k(J), \qquad \mathrm{Lead}_{\lambda\mu}(0) = (-1)^m \#\{H\} \neq 0,$$

where a (tight) history processes $\lambda_\ell, \dots, \lambda_1$, each part k either opening a block of size k or merging a set of $p \ge 1$ existing (distinguishable) blocks of sizes $J = (j_1, \dots, j_p)$ into one block of size $k + |J|$ via the order-p piece $E^{(p)}_k$, it ends at μ if the final block sizes are μ, and $W_k(J) :=$ $\mathrm{lin}_e\, [\cdots [E^{(p)}_k, e_{j_1}], \dots, e_{j_p}](1)$.

Proof idea. Pass to $\mathrm{gr}_I \Lambda$: non-tight terms land in $I^{\ell-m+1}$; tight steps are multilinear derivations, so only linear parts $g_i \equiv \gamma_i e_{n_i} \bmod I^2$ matter. At $t = 0$, a roots-of-unity evaluation, Weyl orthogonality and Murnaghan–Nakayama give $W_k(J)|_{t=0} = (-1)^p$. □

Theorem 3.8 (Merge weights)**.** *With $n = k + |J|$ and $p = |J|$,*

$$W_k(J) = (-1)^p \frac{[n]_t}{[k]_t} \prod_{j \in J} [k]_{t^j}.$$

Consequently $(-1)^m \mathrm{Lead}_{\lambda\mu}(t) > 0$ for coarsenings and every real $t > 0$.

Proof idea. $W_k(J) = (-1)^{d-p} \mathrm{lin}_e\, T_k(p_J)$, $d = |J|$, with T_k as in Lemma 3.9. The parabolic factorisation $T_k P_\rho = P_{\rho + 1^k}$ [13, III (2.2)] and $\mathrm{lin}_e\, P_\lambda = (-1)^{n-k} \frac{[n]}{[k]} P_{\lambda - 1^k}(1, t, \dots, t^{k-1})$ (from [13, III.7 Ex. 2, III.2 Ex. 1]) give the "discrete HL measure = principal specialisation" identity $\mathrm{lin}_e\, T_k f = (-1)^d \frac{[n]}{[k]} f(1, t, \dots, t^{k-1})$ (Lemma 3.9); apply it to $f = p_J$. □

Lemma 3.9. *Let $T_k f := \sum_{|A|=k} c_A X_A f(X_A)$ for $f \in \Lambda_k$. For f homogeneous of degree d and $n = k + d$:*

$$\operatorname{lin}_e T_k(f) = (-1)^d \frac{[n]_t}{[k]_t} f(1, t, \dots, t^{k-1}).$$

Remark 3.10 (Novelty). Lemma 3.9 follows from [13, III.7 Ex. 2, III (2.2)]; probably known.

4 Leading coefficients: graph cumulants and blocks

Theorem 4.1 (Full-merge lead)**.** *For $\lambda \vdash n$ with $\ell = \ell(\lambda) \geq 2$,*

$$\operatorname{Lead}_{\lambda,(n)}(t) = \frac{1 - t^n}{\prod_i (1 - t^{\lambda_i})} K_\lambda(t), \qquad K_\lambda(t) := \sum_{H \text{ connected on } [\ell]} \prod_{ij \in H} (t^{\lambda_i \lambda_j} - 1).$$

K_λ is the connected part (joint cumulant) of $t^{e_2(\lambda)} = \prod_{i<j}(1 + (t^{\lambda_i \lambda_j} - 1))$.

Proof idea. (1) Tight histories ending in one block are increasing trees on $[\ell]$ (parent = the merging part). (2) With N_v the λ-mass of the subtree at v, Theorem 3.8 telescopes along the tree to $\frac{1-t^n}{\prod(1-t^{\lambda_i})} \prod_{v \to c} (t^{\lambda_v N_c} - 1)$. (3) Tree–graph identity: $\sum_{\text{incr. trees}} \prod_{v \to c} (\prod_{j \in T_c} x_{vj} - 1) = \sum_{\text{conn. } H} \prod_{e \in H} (x_e - 1)$ for generic x, by the minimum-vertex/component decomposition; specialise $x_{ij} = t^{\lambda_i \lambda_j}$. □

Corollary 4.2. *(a) $\lambda = 1^n$:* $\operatorname{Lead} = (-1)^{n-1}[n]_t \, I_n(t)$ *with I_n the tree inversion enumerator [14]. (b) $t = 0$:* $\operatorname{Lead} = (-1)^{\ell-1}(\ell - 1)!$. *(c) $t \to 1$:* $\operatorname{Lead} \to (-1)^{\ell-1} n^{\ell-1}$ *(weighted Cayley). (d)* $\operatorname{sgn} \operatorname{Lead} = (-1)^{\ell-1}$ *for all $t > 0$.*

Proof. At $t = 0$ every edge weight is -1 and $\sum_{H \text{ conn.}} (-1)^{|H|} = (-1)^{\ell-1}(\ell - 1)!$. As $t \to 1$, with $\varepsilon = t - 1$, only spanning trees contribute to the $\varepsilon^{\ell-1}$ term of K_λ, and $\sum_T \prod_{ij \in T} \lambda_i \lambda_j = (\prod_i \lambda_i) \, n^{\ell-2}$ (weighted Cayley); the prefactor is $(-n\varepsilon)/((-\varepsilon)^\ell \prod \lambda_i) \cdot (1 + O(\varepsilon))$. (d) is the case $\mu = (n)$, $m = \ell - 1$, of the sign statement in Theorem 3.8: each merge contributes $(-1)^p$ times a positive quantity, $\sum p = m$ along every history, and histories exist. □

Remark 4.3 (Prior art — graph half). The tree, connected-graph and cumulant forms of K_λ are a special case of Dołęga [5, Prop. 2.1 and (11), Lemma 2.3] ($K_\lambda = (t-1)^{\ell-1} T(1, t)$ for the Tutte polynomial of the multigraph with $\lambda_i \lambda_j$ edges between i and j), cf. also [6, 14]; step (3) is a Penrose-type tree–graph identity [16]. Ours: the identification of the $\star$-lead with prefactor$\cdot K_\lambda$ (Theorems 3.7, 3.8 and 4.1) and Theorems 4.4 and 4.5. Haglund–Tewari's single-row cumulant [8, Def. 7.1, Thm. 7.3] (their q is our t) is the same graph half: since $\langle \widetilde{H}_{(m)}, e_m \rangle = q^{\binom{m}{2}}$, their cumulant $\kappa(\lambda)$ (we write C_λ) satisfies $\langle C_\lambda, e_n \rangle = t^{n(\lambda')}(t-1)^{1-\ell} K_\lambda(t)$. *Separator.* Full-merge leads $L(\lambda)$ of any such theory are only defined up to the gauge $L(\lambda) \mapsto L(\lambda) \, a^{\ell(\lambda)} g(|\lambda|) \prod_i f(\lambda_i)$ (rescaling generators,

degree-wise normalisation, rescaling the expansion parameter). The first gauge invariant appears at $n = 4$: $J := L(1^4)L(2,2)/L(2,1,1)^2$ (the exponents of a, $g(4)$, $f(1)$ and $f(2)$ all cancel). For $\star$, Theorem 4.1 gives $J = \frac{(t^2+1)(t^3+3t^2+6t+6)}{(t+1)(t^2+t+2)^2}$, and Haglund–Tewari's $\langle C_\lambda, e_n\rangle$ gives the same value. Dołęga's $\oplus$-cumulant of columns (lowest $(q-1)$-coefficient of $[g_n]\widetilde{H}_{\lambda'}$, $g_k = \widetilde{H}_{1^k}$) gives, by computation, $J \equiv 3/2 = J_\star(0)$, whereas $J_\star(1) = 1$, the Cayley value. So up to gauge Dołęga's $\oplus$ is t-independent and agrees with $\star$ only at $t = 0$, where the leads are partition-lattice Möbius values (Corollary 4.2(b)), while Haglund–Tewari's is the same graph object without $\star$.

Theorem 4.4 (Coarsening leads factor)**.** *For μ a coarsening of λ,*

$$\mathrm{Lead}_{\lambda\mu} = \sum_{\pi} \prod_{B\in\pi} \mathrm{Lead}_{\lambda_B,(|\lambda_B|)},$$

summed over set partitions π of the positions $[\ell]$ whose sorted block sums equal μ (singletons contribute 1). There is no interleaving factor.

Proof idea. Increasing forests = set partition × increasing tree per block, and the telescoped weight is local to each tree. □

Theorem 4.5 (Block multiplicativity)**.** *Let $\kappa = \kappa(\lambda,\mu) \geq 1$, $\ell = \ell(\lambda)$, $m = \ell - \kappa$. Then*

$$[(s-1)^m]\, c_{\lambda\mu} = \sum_{\pi} \sum_{(\nu^C)_{C\in\pi}} \prod_{C\in\pi} [(s-1)^{|C|-1}]\, c_{\lambda_C,\nu^C},$$

where π runs over set partitions of $[\ell]$ into exactly κ blocks and (ν^C) over tuples with $\bigsqcup_C \nu^C = \mu$ and $\nu^C \trianglerighteq \lambda_C$; every factor then has $\kappa(\lambda_C,\nu^C) = 1$. With Theorem 3.4 this is $\mathrm{Lead}_{\lambda\mu}$.

Proof idea. Expand by polarization into histories; non-tight histories produce $> \kappa$ compatible blocks and vanish by maximality of κ; tight histories factor block by block and interleave uniquely. Commutativity of $\star$ passes from compositions to partitions. □

Remark 4.6. Theorem 4.4 is the case where each ν^C is a single part. The factorisation has the shape of a Mayer (cluster) expansion, i.e. moment–cumulant inversion of the graph identity in Theorem 4.1; the $\star$-specific content is that the connected leads are those of Theorem 4.1, with no interleaving correction.

5 Box Complement and the linear coefficient

Theorem 5.1 (Box Complement)**.** *Let λ, μ have at most ℓ parts (padded with zeros) and all parts $\leq N$; put $N^\ell - \lambda := (N-\lambda_\ell, \ldots, N-\lambda_1)$. Then*

$$c_{N^\ell-\lambda,\, N^\ell-\mu}(s,t) = s^{N\binom{\ell}{2}-(\ell-1)|\lambda|}\, c_{\lambda\mu}(s,t).$$

Proof idea. Stability of the subset formula for every N (also $N <$ degree), and an inversion lemma in N variables: for $F = e_N^m G(1/x)$ with G of degree g,

$$E_{N-k}F = s^{m(N-k)-g} e_N^{m+1} (E_k G)(1/x)$$

(substitute $y = 1/x$, $B = A^c$). Iterate, and compare coefficients using the identity $e_{N-r}(x) = e_N(x)\, e_r(1/x)$ and Theorem 2.2 (support has $\le \ell$ parts). □

Corollary 5.2 (Column Lemma; lead invariance). $c_{\lambda+1^\ell,\,\mu+1^\ell} = s^{\binom{\ell}{2}} c_{\lambda\mu}$. *Hence* v *and* Lead *are invariant under* $(\lambda,\mu) \mapsto (\lambda+1^\ell, \mu+1^\ell)$ *and* $(\lambda,\mu) \mapsto (N^\ell-\lambda, N^\ell-\mu)$; *e.g.* $(2,2,2) \to (3,3)$ *and* $(2,2,2) \to (4,1,1)$ *have the lead* $[3]_t(t+2)$ *of* $(1,1,1) \to (3)$.

Remark 5.3 (Prior art). The inversion lemma follows in a few lines from [4, Rem. 3.3] ($S\Gamma_I S = \Gamma_I^{-1}$ under $x \mapsto x^{-1}$) plus subset complement; stability is [9, Prop. 3.8, Cor. 3.10]. We have not found Theorem 5.1 or Corollary 5.2 for $\star$ in the literature.

Theorem 5.4 (Plethystic linear coefficient). *For* G *homogeneous of degree* d: $\mathrm{lin}_e(e_k \star G) = (-1)^d \frac{[k+d]_t}{[k]_t}\, G[(s-1)[k]_t]$, *where* $G \mapsto G[(s-1)[k]_t]$ *is the ring map* $p_r \mapsto (s^r-1)[k]_{t^r}$.

Corollary 5.5 (Pieri rule, all s). $e_k \star e_r = \sum_{y=0}^{\min(k,r)} s^y C_{k-y,r-y}\, e_{k+r-y} e_y$ *with* $C_{a,0} = C_{0,b} = 1$, $C_{a,b} = (-1)^b \frac{[a+b]}{[a]} \pi_a(b)$, $\pi_a(j) = [u^j] \prod_{m=0}^{a-1} \frac{1+sut^m}{1+ut^m}$.

Proof idea. Support by Theorem 2.2; strip y columns by Corollary 5.2 (valid for compositions) to reduce to $\mathrm{lin}_e\, E_{k-y}(e_{r-y})$; evaluate by Theorem 5.4, which itself is Lemma 3.9 applied to the generating function $E_k(\prod_i E(u_i)) = \prod_i E(u_i)\, T_k(\prod_i \prod_{a\in A} \frac{1+su_ix_a}{1+u_ix_a})$. □

Remark 5.6. Corollary 5.5 was first obtained by a different route, a k-fold kernel computation; Hikita's [9, Thm. 3.12] is the case $k = 1$. Explicit rules for $e_k \star e_\lambda$ with λ of two or ℓ columns will appear in the long version.

6 Second order: every valuation-2 lead

By Theorem 4.5, valuation-2 leads reduce to connected leads with $\ell(\lambda) = 3$, $\kappa = 1$; by Corollary 5.2 to two-part μ. With T_a as in Lemma 3.9, put

$$\Gamma_a(e_b, e_c) := E_a^{(2)}(e_b e_c) - e_b E_a^{(2)}(e_c) - e_c E_a^{(2)}(e_b) = \sum_{1\le r\le b,\ 1\le q\le c} (-1)^{r+q} e_{b-r} e_{c-q}\, T_a(p_r p_q).$$

Proposition 6.1. *If* (a,b,c) *is any ordering of* λ *and* $\kappa(\lambda,\mu) = 1$, *then* $[(s-1)^2] c_{\lambda\mu} = [e_\mu]\big(\Gamma_a(e_b,e_c) + D_a D_b(e_c)\big)$.

Proof idea. By commutativity and associativity of $\star$, $e^\star_\lambda = E_a E_b E_c(1)$ for any ordering; expand by (2.1). As $E^{(r)}_c(1) = 0$ for $r \geq 1$, the terms are $E^{(2)}_a(e_b e_c)$, $e_a E^{(2)}_b(e_c)$, $D_a D_b(e_c)$; the terms to discard, $e_a E^{(2)}_b(e_c)$, $e_b E^{(2)}_a(e_c)$, $e_c E^{(2)}_a(e_b)$, are one of these plus the two subtracted in Γ_a (the list is complete). Each has $\kappa \geq 2$: $E^{(2)}_a(e_c) = [(s-1)^2] e^\star_{(a,c)}$ lives on e_ν, $\nu \trianglerighteq (a,c)$ (Theorem 2.2), with b its own block; likewise for the others. □

Theorem 6.2 (Constant-term adjoint formula)**.** *For $g \in \Lambda_a$, $F \in \Lambda$: $\varphi_a(t)\,\langle T_a g, F\rangle_t = \mathrm{CT}\big[Z\, g(z)\, F(z_1^{-1}, \dots, z_a^{-1})\, \Omega(z)\big]$, where $Z = z_1 \cdots z_a$ and $\Omega(z) = \prod_{1 \leq i < j \leq a} \frac{z_j - z_i}{z_j - t z_i}$, expanded in z_i / z_j $(i < j)$.*

Proof idea. $T_a g$ restricted to a variables is Zg and lies in $\mathrm{span}\{P_\mu : \ell(\mu) = a\}$. Evaluate $\langle G, q_\alpha\rangle_t = [y^\alpha] G$ via Cauchy [13, III (4.2), (4.4), (4.8)–(4.9)], write $\langle G, Q_\lambda\rangle_t$ as a constant term by the raising-operator formula [13, III (2.15′)], and symmetrise $\Omega = \Delta\theta$ with $\sum_w w(\theta) = v_a(t)$ [13, III (1.4)]; $Q_\lambda(z^{-1})$ is evaluated by [13, III (2.1), (2.11)–(2.12)]. This is Jing's vertex-operator calculus [10] with the vertex operators hidden. □

Theorem 6.3 (Two-point formula)**.** *Let $g \in \Lambda_a$ be homogeneous of degree d, $n = a + d$, $x, y \geq 1$, $x + y = n$, and $\Phi_a(g; x, y) := \langle T_a g, p_x p_y\rangle$ (Hall pairing, not $\langle\cdot,\cdot\rangle_t$). For $1 \leq A \leq a-1$ put $B = a - A$ and $G_A(w) = \sum_k G_{A,k} w^k := g(1, t, \dots, t^{A-1}, w, wt, \dots, wt^{B-1})$. Then*

$$\Phi_a(g; x, y) = (-1)^a (1 - t^x)(1 - t^y) \Bigg\{ \sum_{A=1}^{a-1} t^{-AB} \Big[\frac{G_{A, y-B}}{(1 - t^A)(1 - t^B)} + \frac{1}{1 - t^a} \sum_{m \geq 1} (t^{-Am} - t^{Bm})\, G_{A, y-B-m} \Big] - \frac{g(1, t, \dots, t^{a-1})}{1 - t^a} \sum_{j=0}^{a-1} t^{-jy} \Bigg\}.$$

Proof idea. Insert $S(u) = \sum_i 1/(uz_i - 1)$ into Theorem 6.2 and take iterated residues: every variable lands on a t-string $1/u, t/u, \dots$ or $1/v, t/v, \dots$. A string of length A with base q has weight $(-1)^{A-1} \varphi_{A-1} q^A$, and two strings interact through the shuffle identity

$$\mathrm{Sh}_{A,B}(w) = \begin{bmatrix} A+B \\ A \end{bmatrix}_t t^{-AB}\, \frac{(1-w)(1 - t^{B-A} w)}{(1 - t^{-A} w)(1 - t^B w)}, \qquad w = u/v,$$

proved by a last-letter recursion. Take Taylor coefficients at $w = 0$. □

Remark 6.4 (Relation to Green polynomials). Write $p_\mu = \sum_\lambda \mathcal{X}^\lambda_\mu(t) P_\lambda(x; t)$. For $g = P_\rho$ and $\lambda = \rho + 1^a$ one has $\Phi_a(P_\rho; x, y) = (1 - t^x)(1 - t^y) \mathcal{X}^\lambda_{(x,y)}(t)/b_\lambda(t)$, so Theorem 6.3 evaluates Green polynomials at two-part classes; the two t-strings are the Frobenius orbits of a maximal torus of type (x, y) [7]. Theorem 6.2 is Jing's constant-term expression

for the matrix element $\langle Q_\lambda, p_\mu \rangle$ [10], cf. [11, (2.17), (2.20)]. Jing and Liu [11, Thm. 2.7] expand it as a nested sum valid for all μ, extracting one variable at a time; at two-part μ its inner sums run over classes of arbitrary length. Their closed forms [11, Thm. 2.10, (2.37)–(2.40)] restrict the Hall–Littlewood index λ, not the class μ; their Murnaghan–Nakayama rule [11, Thm. 3.2] generalises a formula of Morris [15][1], its case $\ell(\lambda) = 2$. Our formula is explicit, with no sum over classes, and linear in the two-string specialisation G_A of g; for general ρ, G_A is itself a finite (branching) sum, so Theorem 6.3 is not a product formula.

Example 6.5 (Two rows)**.** For $a = 2$, only $A = B = 1$ occurs, and $G_1(w) = P_\rho(1, w; t) = w^{\rho_2} \frac{(1-tw)-w^r(w-t)}{1-w}$ with $r = \rho_1 - \rho_2 \geq 1$ ($G_1 = w^{\rho_2}$ if $r = 0$). Substituting into Theorem 6.3 gives, for $\lambda = (\lambda_1, \lambda_2)$, $y \leq x$, and $m_{xy} = 2$ if $x = y$, 1 otherwise,

$$\mathcal{X}^{(\lambda_1,\lambda_2)}_{(x,y)}(t) = \begin{cases} (t-1)\, t^{\lambda_2-1-y}(1+t^y), & y < \lambda_2, \\ m_{xy} - (1-t)\, t^{\lambda_2-1}, & y = \lambda_2, \\ (t-1)\, t^{\lambda_2-1}, & y > \lambda_2, \end{cases}$$

in agreement with a Kostka–Foulkes computation for $|\lambda| \leq 10$. This is not new: for $\lambda = (n-k, k)$ and every class ρ, Jing and Liu [11, Thm. 2.6, unnumbered display after (2.32), same in v1 and v2] give $\mathcal{X}^\lambda_\rho = \pi_k + (t-1)\sum_{j<k} t^{k-1-j}\pi_j$, with π_j the number of j-subsets fixed by ρ (their $D^{(j)}(\rho)$). A short proof: $\mathcal{X}^\lambda_\rho = \sum_\nu \chi^\nu_\rho K_{\nu\lambda}(t)$ [13, III (7.6′)], where $K_{(n-j,j),\lambda} = t^{k-j}$ (one tableau, and K is monic of degree $n(\lambda) - n(\nu)$ [13, III (6.5)]) and $\chi^{(n-j,j)}_\rho = \pi_j - \pi_{j-1}$ by Young's rule. On the diagonal class $(x, y) = \lambda$ it gives $t^{\lambda_2} - t^{\lambda_2-1} + m_{xy}$, which for $\lambda_1 > \lambda_2$ with $\lambda_2 \geq 3$ odd equals -1 at $t = -1$ and 1 at $t = 0$. A root in $(-1, 0)$ rules out any product or quotient of cyclotomic factors and powers of t, so Green polynomials at two-part classes admit no product formula even for two rows; for expressions $t^c \prod_i (1 - t^{d_i})^{e_i}$ with $e_i \in \mathbb{Z}$ this was observed independently in [1, Thm. D].

Theorem 6.6 (Closed $\ell = 3$, $\kappa = 1$ leads)**.** *Let $\lambda = (a, b, c)$ be any ordering of a length-3 partition, $\mu = (x, y)$ with $\kappa(\lambda, \mu) = 1$, $n = x + y$, $m_{xy} = 2$ if $x = y$ and 1 otherwise. Put $\Xi_a(r, q) := \lim_e T_a(p_r p_q) = (-1)^{r+q}[a + r + q]\,[a]_{t^r}[a]_{t^q}/[a]$ and $U_a(b, c; x, y) := \big((-1)^n \Phi_a(p_b p_c; x, y) - \Xi_a(b, c)\big)/m_{xy}$. Then*

$$[(s-1)^2]\, c_{\lambda\mu} = (-1)^{b+c} U_a(b, c; x, y) + \sum_{\substack{1 \leq r < b \\ \{b-r,\, a+r+c\} = \{x,y\}}} (-1)^{r+c} \Xi_a(r, c) + \sum_{\substack{1 \leq q < c \\ \{c-q,\, a+b+q\} = \{x,y\}}} (-1)^{b+q} \Xi_a(b, q) + [e_x e_y]\, D_a(M_{bc}),$$

[1] We have not seen [15] first-hand; its scope is as described in [11, p. 12, before Thm. 3.2].

with D_a the derivation $D_a(e_j) = M_{aj}$ of Proposition 3.3.

Proof idea. Proposition 6.1; in the expansion of Γ_a only terms with $r = b$ or $q = c$ survive $[e_x e_y]$; single-e parts come from Lemma 3.9, and $\langle G, p_x p_y \rangle = (-1)^n (m_{xy}[e_x e_y]G + \mathrm{lin}_e\, G)$ converts Theorem 6.3 into $U_a = [e_x e_y]\, T_a(p_b p_c)$. □

Corollary 6.7. *Every lead at valuation 2 ($\kappa = \ell - 2$, any ℓ) has a closed formula: by Theorem 4.5 it is a sum of products of pair leads (Proposition 3.3) and $\ell = 3$, $\kappa = 1$ leads (Theorem 4.1, Corollary 5.2, Theorem 6.6).*

Example 6.8. $(3,3,3) \to (7,2)$: Lead $= 2t^{13} + 3t^{12} + 3t^{11} + 6t^{10} + 6t^9 + 6t^8 + 9t^7 + 6t^6 + 3t^5 + 9t^4 + 6t^3 + 3t + 4$. Leads are *not* positive in general: $(4,4,2) \to (7,3)$ has Lead $= (t+1)(t^2+1)(2t^8 + t^7 + t^6 + 3t^4 + t^3 + t^2 - t + 2)$.

Computer checks. All statements in this section were checked against an independent implementation of Hikita's subset formula: Theorem 6.6 on 16 pairs with $n \le 12$ (both examples among them) and on 27 pairs in all 3 orderings with $n \le 10$; Theorem 6.3 in 229 cases ($a \le 5$); and the Green-polynomial dictionary symbolically for $|\lambda| \le 8$ and numerically for $|\lambda| \le 9$.

7 Open problems

1. **Valuation ≥ 3.** Connected leads with $\ell \ge 4$, $\kappa = 1$. Hierarchy: $v = 1 \leftrightarrow p_n \leftrightarrow$ one t-string (Lemma 3.9); $v = 2 \leftrightarrow p_x p_y \leftrightarrow$ two strings (Theorem 6.3). Does the three-string shuffle sum factor pairwise?

2. **Zero set at generic t.** At $t = -2$ seven pairs with $n = 6$ have $v > \ell - \kappa$; for coarsenings this is cancellation between histories, e.g. $\mathrm{Lead}_{(1,1,1),(3)} = [3]_t(2+t)$. Describe the zero set of $[(s-1)^{\ell-\kappa}]c_{\lambda\mu}$ in t.

3. **Counting.** A formula for $N(\lambda, \mu)$ in Theorem 3.4; for $\ell(\lambda) = 3$, $\mu = (n-1, 1)$, $\kappa = 1$ we observe $\mathrm{Lead}_{\lambda\mu}(1) = 2n^2 - 6n + 3$, independent of λ ($n \le 10$, computed).

4. **Positivity.** Leads are not t-positive (Example 6.8). Is there a signed or cancellation-free model?

5. **Cumulant link.** Is there a Macdonald-cumulant construction whose leads are the $\star$-leads for all t, i.e. cumulants of $e_k \mapsto t^{\binom{k}{2}}$?

Acknowledgements

[TODO: Robin; Clio (peer agent) for reviews; per Robin.]

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AI declaration

[TODO: Robin: edit wording and fill in your role below. Mandatory, uncapped, not counted toward the 12 pages.]

The mathematics in this extended abstract was produced by an AI agent. "Rick" is an autonomous AI research agent built on Anthropic's Claude models (Claude Opus 5.5 in the final sessions; earlier sessions ran on earlier Claude versions **[TODO:** Robin: confirm the list]). It works in a sandbox with Python/SymPy, LaTeX, and arXiv/Semantic Scholar search, and keeps a written log of every session. Rick chose the questions, ran the computer experiments, conjectured the statements, found and wrote the proofs, did the literature and novelty searches, and drafted this text.

- **Statements and proofs.** Every result not attributed to a reference was conjectured and proved by the agent: the dominance support, the biderivation and the matrix M, the block valuation law, the merge-history expansion and merge weights, the full-merge, coarsening and block-multiplicative lead formulas, the Box Complement theorem and Column Lemma, the plethystic linear coefficient and Pieri rule, the constant-term and two-point formulas, and the valuation-2 leads. Each proof was written in full in the agent's notes; the merge weights and the second-order results (Section 6) were in addition re-derived from scratch in a later session. The proofs sketched here are condensed from those notes.

- **Computer verification.** All theorems were checked in small cases by the agent's own Python and SymPy code against a direct implementation of Hikita's subset formula. Sub-agents (further instances of the same model) were used for some computations and for drafting parts of the LaTeX; their outputs were checked by the agent.

- **Review by a second AI agent.** A separate AI research agent ("Clio", also Claude-based, run independently) reviewed some inputs, but not all: the subset-formula inputs; the $s \to 0$ identification and the location of the matrix d in the literature (Remark 2.3); the structure of the two proofs of Theorem 3.8 (they share one lemma, so they are not independent); an independent check of the dominance support for $n \leq 7$; a numerical cross-check of the Green-polynomial dictionary after Theorem 6.3; Proposition 6.1 (proof idea read; statement checked against Clio's own implementation of Proposition 2.1); and Theorem 6.6 (read, and cross-checked numerically against that implementation at $\lambda = (2,2,2)$). The block valuation law, Theorems 4.1, 4.4, 4.5 and 5.1, Theorems 6.2 and 6.3 have not been reviewed by another agent (Theorem 6.3 was spot-checked numerically only).

- **Literature.** Prior-art searches were done by the agent and its sub-agents. Wherever a result or mechanism turned out to be known, it is cited and not claimed (e.g.

Remark 2.3, the remarks after Theorems 4.1 and 5.1). Textbook equation numbers in [13] were checked against the book.

- **Human role.** **[TODO:** Robin: describe your role (e.g. supervision, submission, any human verification). As of this draft no proof has been fully checked by a human.**]**